# Delta Physics

Select a topic of your Choice

5 minutes – Individual

# Gravitation

Why does the Moon take a month to go round?



Higher Physics

PART 3

# Beyond Higher and Exam Practice

The Space Station goes round the Earth in 90 minutes. Why does the Moon take a month?

# By the end of this part…



1. I can **use g = GM/r²** (beyond Higher)
2. I can **describe** how **g** changes with **height**
3. I can **explain** why **higher orbits** are **slower**
4. I can **answer** exam questions on **$F = Gm_1m_2/r^2$**

# Where g Comes From

Beyond Higher — not in the exam

## Weight

$W = mg$

## Newton's law

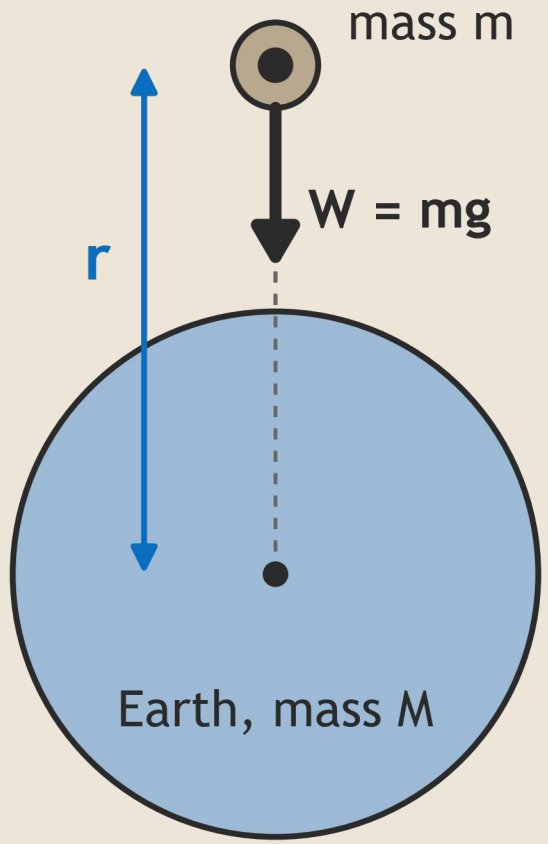
$W = GMm/r^2$

## So

$mg = GMm/r^2$: **m cancels**

## Result

$g = GM/r^2$




r: from the centre

# Field Strength g

**Beyond Higher — not in the exam**

$$g = (G\,M)/r^2$$

**g**

field strength (N kg$^{-1}$)

**G**

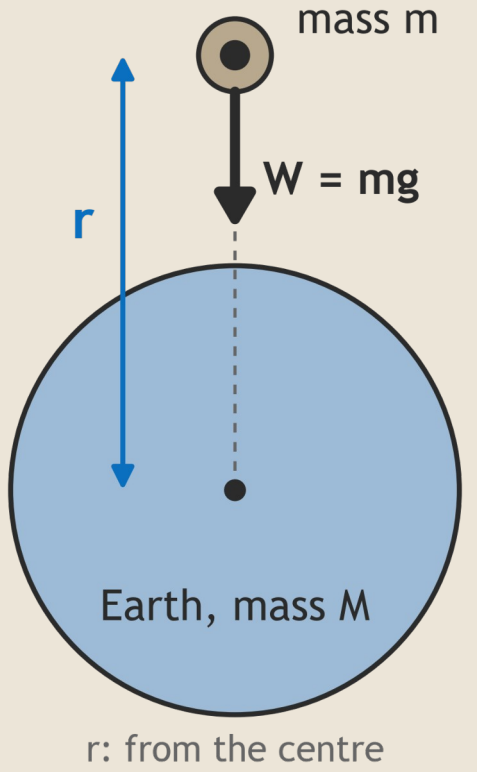
on the data sheet

**M**

planet's mass (kg)

**r**

from the centre (m)



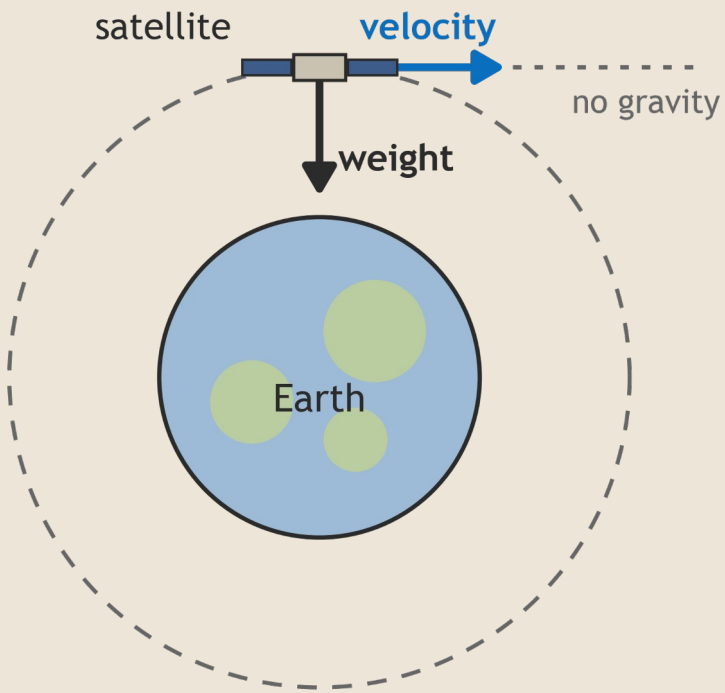
# Example: g at the ISS

**Beyond Higher — not in the exam**

The ISS orbits $6.82 \times 10^6$ m from the centre of the Earth.

The mass of the Earth is $6.0 \times 10^{24}$ kg.

**Calculate** the gravitational field strength at the ISS. **3**



# Example: Marking

Beyond Higher — not in the exam



| Question | Expected response | Max mark | Additional guidance |
|---|---|---|---|
| | $g = GM/r^2$ (1 mark)<br>$g = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} / (6.82 \times 10^{6})^2$ (1 mark)<br>$g = 8.6$ N kg$^{-1}$ (1 mark) | 3 | About 90 % of 9.8 N kg$^{-1}$: gravity has not gone (Part 1) |

# g Against Distance

**Beyond Higher — not in the exam**

## Surface

**9.8** N kg$^{-1}$

## ISS

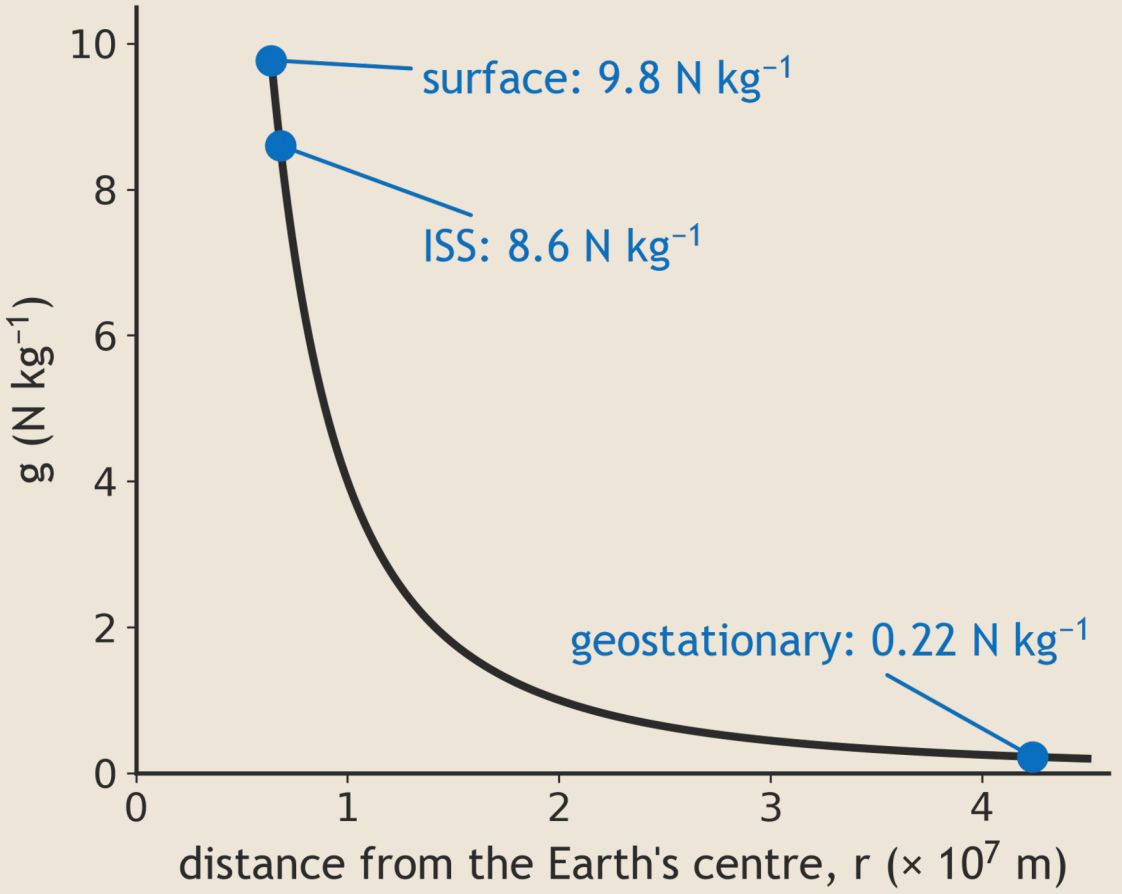
**8.6** N kg$^{-1}$

## Geostationary

**0.22** N kg$^{-1}$

## Double r

**g ÷ 4**: inverse square



# Orbital Speed

**Beyond Higher — not in the exam**

$$v = \sqrt{((G\,M)/r)}$$

**v**
orbital speed (m $s^{-1}$)

**G**
on the data sheet

**M**
planet's mass (kg)

**r**
orbit radius (m)



# Orbital Period

**Beyond Higher — not in the exam**

| | ISS | Geostationary |
|---|---|---|
| r (m) | $6.82 \times 10^6$ | $4.24 \times 10^7$ |
| v ($m\ s^{-1}$) | $7.7 \times 10^3$ | $3.1 \times 10^3$ |
| T | $5.6 \times 10^3$ s = 93 min | $8.7 \times 10^4$ s = 24 h |

## Speed

$v = \sqrt{(GM/r)}$

## One lap

$T = 2\pi r/v$

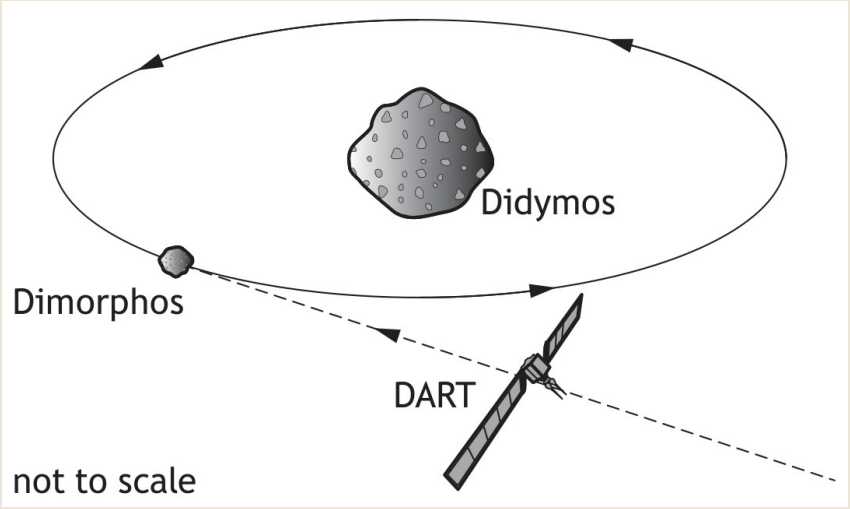
## 24 h

keeps pace with the Earth: **dishes do not move**

# Past Paper: 2024 Paper 2 Q4(a)(ii)

4. (a) The mass of Didymos is taken to be $5.3 \times 10^{11}$ kg.
The mass of the Earth is $6.0 \times 10^{24}$ kg.

(ii) When DART collided with Dimorphos, the distance between Earth and Didymos was $1.1 \times 10^{10}$ m.
Determine the gravitational force between Earth and Didymos.




3

**Scan**: page 15.

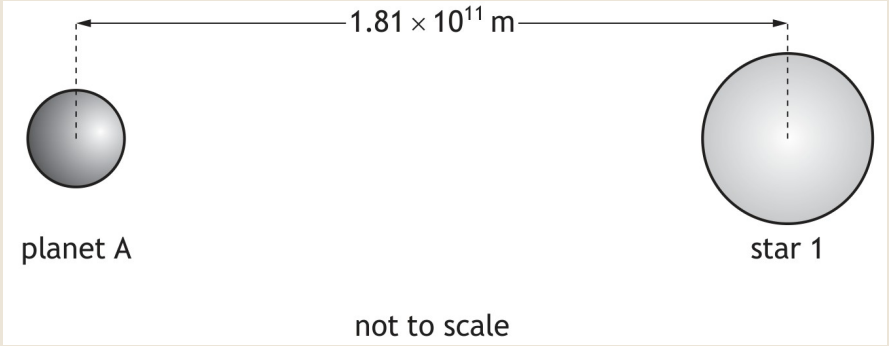
# Marking: 2024 Paper 2 Q4(a)(ii)

| Question | Expected response | Max mark | Additional guidance |
|---|---|---|---|
| **4 (a)(ii)** | $F = Gm_1m_2/r^2$ (1 mark)<br>$F = 6.67 \times 10^{-11} \times 5.3 \times 10^{11} \times 6.0 \times 10^{24} / (1.1 \times 10^{10})^2$ (1 mark)<br>$F = 1.8 \times 10^{6}$ N (1 mark) | 3 | Accept: 2, 1.75, 1.753 |

# Past Paper: 2025 Paper 2 Q5(c)(i)

**5.** (c)

(i) One exoplanet, planet A, has a mean orbital radius of $1.81 \times 10^{11}$ m and a mass of $6.09 \times 10^{24}$ kg.
The mass of star 1 is $1.77 \times 10^{29}$ kg.
Calculate the gravitational force between planet A and star 1.




**3**

**Scan**: page 18.

# Marking: 2025 Paper 2 Q5(c)(i)

| Question | Expected response | Max mark | Additional guidance |
|---|---|---|---|
| **5 (c)(i)** | $F = Gm_1m_2/r^2$ (1 mark)<br>$F = 6.67 \times 10^{-11} \times 6.09 \times 10^{24} \times 1.77 \times 10^{29} / (1.81 \times 10^{11})^2$ (1 mark)<br>$F = 2.19 \times 10^{21}$ N (1 mark) | 3 | Accept: 2.2, 2.195, 2.1946 |

# Past Paper: 2025 Paper 2 Q5(c)(ii)

**5.** (c) (continued)

(ii) A second exoplanet, planet B, has three times the mass of planet A.
The mean orbital radius of planet B is double the mean orbital radius of planet A.
State how the gravitational force of attraction between planet B and star 1 compares to the gravitational force of attraction between planet A and star 1.
Justify your answer.

**3**

# Marking: 2025 Paper 2 Q5(c)(ii)

| Question | Expected response | Max mark | Additional guidance |
|---|---|---|---|
| **5 (c)(ii)** | Force is less/¾ (1 mark)<br><br>Three times the mass gives three times the force. (1 mark)<br><br>Doubling the distance gives quarter of the force. (1 mark) | 3 | JUSTIFY<br>Look for this statement first - if incorrect or missing then 0 marks.<br>Do not accept: decrease by ¾<br>Can justify by calculation:<br>$1.65 \times 10^{21}$ N<br>Accept: 1.6, 1.646, 1.6460 |

# What Candidates Get Wrong

## Substituting

square r; use G, not g (2024 Q4(a)(ii))

## Statement first

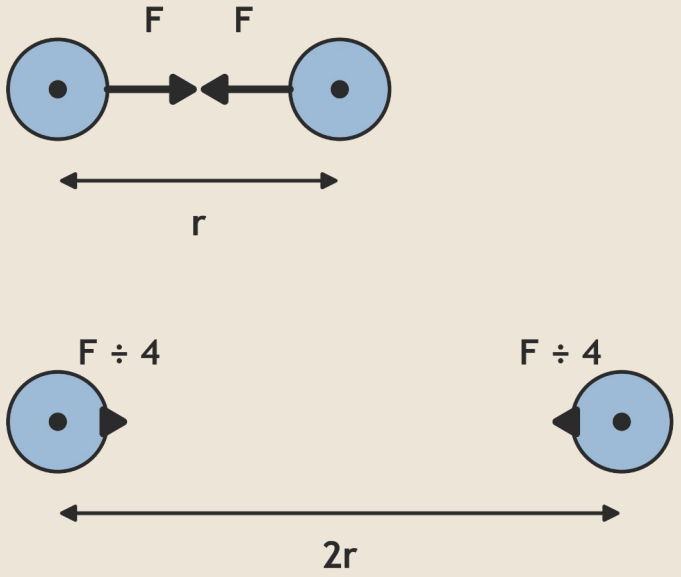
or 0 marks (2025 Q5(c)(ii))

## Justify

only some did (2025 Q5(c)(ii), 46 %)

## Wording

'decrease by ¾': 0 (2025 MI)



# Look back — by the end of this part…



1. I can **use g = GM/r²** (beyond Higher)
2. I can **describe** how **g** changes with **height**
3. I can **explain** why **higher orbits** are **slower**
4. I can **answer** exam questions on **$F = Gm_1m_2/r^2$**

# Course specification — 1 of 1

- Knowledge that satellites are in free fall around a planet/star.
- Use of Newton's Law of Universal Gravitation to solve problems involving force, masses and their separation.

**Scan**: the course specification, page 8.

# Credits

**Title: the ISS:** NASA

**Past Paper: 2024 Paper 2 Q4(a)(ii):** © Qualifications Scotland (SQA). 2024 Higher Physics Paper 2, Q4

**Marking: 2024 Paper 2 Q4(a)(ii):** © Qualifications Scotland (SQA). 2024 Higher Physics Paper 2, Q4

**Past Paper: 2025 Paper 2 Q5(c)(i):** © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

**Marking: 2025 Paper 2 Q5(c)(i):** © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

**Past Paper: 2025 Paper 2 Q5(c)(ii):** © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

**Marking: 2025 Paper 2 Q5(c)(ii):** © Qualifications Scotland (SQA). 2025 Higher Physics Paper 2, Q5

**Course specification — 1 of 1:** © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8

R Stewart - November 2026